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From point-to-point to hub-and-cluster: a systematic review of co-optimization of source–sink matching and transport networks for large-scale CO_2 sequestration

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Large-scale carbon capture, utilization, and storage (CCUS) requires coordinated source–sink matching (SSM), transport infrastructure, and geological storage planning. This review examines how published studies couple source allocation, network topology and capacity, CO_2 flow, and storage availability. Following PRISMA 2020 guidance, a reproducible Web of Science and Scopus search identified 17 studies for direct evidence synthesis. The evidence shows that integrated models can jointly represent source–sink allocation, transport-network investment, operational flow, and storage constraints, whereas sequential approaches may overlook system-wide trade-offs. Mathematical programming remains the most auditable basis for strategic planning because it provides explicit feasibility constraints and, in some cases, optimality bounds. Hub-and-cluster and multimodal configurations can improve infrastructure sharing, but their value depends on utilization, phased connection, access conditions, and cost recovery. Evidence for metaheuristics and AI-enabled methods remains case-specific; their comparative value cannot be established without common instances and complete reporting of feasibility, solution quality, and computational effort. Future research should develop physically valid large-scale benchmarks, adaptive planning models, and governance-aware mechanisms for shared CCUS infrastructure.

KEYWORDS
carbon capture utilization and storage (CCUS), climate change mitigation, environmental systems engineering, hub-and-cluster networks, infrastructure planning, source-sink matching, systems optimization

1 Introduction

As atmospheric greenhouse gas concentrations rise, climate change intensifies, driving increases in global mean temperature. The growing frequency and severity of extreme weather events, coupled with growing risks to natural systems, have made rapid greenhouse gas mitigation and the low-carbon transformation of energy systems urgent imperatives for global climate governance (UNEP, 2025; World Meteorological Organization, 2026). The Paris Agreement mandates holding the increase in global average temperature to well below 2 °C above pre-industrial levels while pursuing efforts to limit it to 1.5 °C, thereby

establishing a clear requirement for deep decarbonization of the global energy system (UNFCCC, 2015). The International Energy Agency (IEA) emphasizes that achieving net-zero emissions by 2050 requires not only accelerating renewable energy deployment, end-use electrification, and energy-efficiency improvements but also the large-scale adoption of low-emission technologies such as carbon capture, utilization, and storage (CCUS). According to the latest IEA CCUS project database, the global CO_2 capture capacity of operational and under-construction projects increased by more than 10% between the first quarter of 2025 and the first quarter of 2026 relative to the previous update, while storage capacity rose by approximately 25% (IEA, 2026a). These trends signal the transition of CCUS from a niche technology to a critical infrastructure component for industrial decarbonization (IEA, 2026b).

CCUS captures CO_2 generated from industrial production and energy use, transporting it to utilization facilities or geological formations for beneficial use or permanent storage, thereby enabling deep emissions reductions across industrial and energy systems. It is increasingly recognized as a critical pathway for advancing the energy transition and achieving global climate goals, particularly in high-emitting sectors such as iron and steel, cement, chemicals, and refining (Fratalocchi, 2026; Global CCS Institute, 2024; Raihan, 2025). In these sectors, the high share of process-related emissions and the limited availability of mature alternatives make deep decarbonization difficult to achieve through fuel switching and energy-efficiency improvements alone, positioning CCUS as an internationally recognized mitigation technology (Hepburn et al., 2019; Jiang et al., 2020). The global CCUS industry has recently entered a phase of accelerated development. In 2025, carbon capture and storage (CCS) projects expanded rapidly worldwide, with the number of operational facilities increasing by 54% year-on-year and 27 new facilities commencing operation. Consequently, shared regional infrastructure and cross-sector cluster deployment are emerging as major directions for industry development (Global CCS Institute, 2025b).

However, CCUS is not a single technology but a complex engineered system encompassing CO_2 capture, compression, transport, utilization, and geological storage. Transport and storage are typically among the most costly components of a CCUS system, while the spatial configuration of emission sources and storage sites directly affects its economic performance, safety, and long-term operational efficiency (Becattini et al., 2022; Middleton and Bielicki, 2009). Large industrial emission sources are concentrated in major energy-producing regions and industrial clusters, whereas suitable geological storage resources, including deep saline aquifers and depleted oil and gas reservoirs, are widely distributed across sedimentary basins. This pronounced spatial mismatch makes CO_2 transport a critical link between capture facilities and storage sites, rendering source-sink matching (SSM) a fundamental prerequisite for CCUS infrastructure planning (Tang et al., 2023; Wei et al., 2021).

Against this background, source-sink matching (SSM) has emerged as a key CCUS infrastructure-planning problem. SSM determines which captured CO_2 volumes are assigned from individual emission sources to eligible storage sinks, subject to capture availability, storage capacity and injectivity, transport connectivity, time, and cost constraints. When source assignments are optimized jointly with candidate corridors, transport capacities, and operating flows, SSM becomes a coupled source-sink-network design problem rather than a distance-based pairing exercise (Wang et al., 2020; Wei et al., 2021; Middleton and Bielicki, 2009).

In recent years, systematic research on CCUS source-sink matching and transport network optimization has expanded worldwide, yielding an integrated research framework spanning resource assessment, system modeling, optimization methods, and intelligent decision-making. Existing studies have mainly addressed the identification of emission sources and storage resources, source-sink matching and CO_2 flow allocation, transport network design, and integrated full-chain optimization. Together, these efforts have established a framework extending from resource assessment to coordinated infrastructure planning (Budiarto et al., 2025; Lin et al., 2026; Promsuk et al., 2025). Methodologically, most studies employ mathematical frameworks such as mixed-integer linear programming (MILP), mixed-integer nonlinear programming (MINLP), network flow models, and multi-period optimization, often in combination with scenario analysis, stochastic programming, robust optimization, and heuristic or metaheuristic algorithms to address large-scale network optimization and planning under uncertainty (Budiarto et al., 2025; Zhang et al., 2025). More recently, emerging approaches such as data-driven modeling, graph learning, and digital twins have been introduced to support network-state prediction, rapid multi-scenario assessment, and dynamic decision support, offering new paradigms and tools for optimizing complex CCUS systems (Mazhar et al., 2026; Raissi et al., 2019; Uddin et al., 2025).

Despite substantial advances, several important gaps remain. First, most studies examine either source–sink matching or transport-network optimization in isolation, with insufficient attention to their interdependence. SSM determines CO_2 transport demand and flow directions, while network topology, capacity, and engineering feasibility feed back into the cost and viability of alternative matches. The two therefore form a coupled infrastructure-planning problem (Fang et al., 2025; Middleton et al., 2012).

Second, the literature remains dominated by static planning models that generally assume fixed emission-source capacities, storage capacities, and transport demands. Long-term dynamics, such as industrial restructuring, carbon-price volatility, declining storage capacity, and policy change, are often inadequately represented. Consequently, these models provide limited support for the long-term and phased deployment of CCUS infrastructure (Bogs et al., 2026; Olson and Yaw, 2025).

Third, the clustering and scaling-up of CCUS projects are making source-sink matching increasingly high-dimensional, multi-objective, and constraint-intensive. Conventional mathematical programming approaches, particularly mixed-integer programming and network flow models, offer strong interpretability and global optimization capabilities. Nevertheless, they continue to face computational challenges in large-scale instances, model scalability, and rapid adaptation to dynamic scenarios (Kattel et al., 2026; Kumcu et al., 2026). Emerging approaches, including artificial intelligence, digital twins, and complex network analysis, have introduced new possibilities for

TABLE 1 Comparison with representative reviews and the verifiable incremental scope of this study.

Review	Primary scope	SSM	Network design	Hubs/ clusters	AI/digital twins	Audit and quantitative evidence
Tapia et al. (2018)	CCUS optimization and decision models	Partial	Partial	Limited	No	Narrative taxonomy
Sleiti et al. (2022)	CO_2 pipeline engineering, safety, cost	No	Yes	Partial	Digital-twin concept	Engineering comparison
Yao et al. (2025)	Offshore storage from a source–sink perspective	Yes	Partial	Partial	No	Domain synthesis
Lin et al. (2026)	Source–sink matching	Yes	Partial	Yes	Partial	Comparative tables
Rahmanta et al. (2026)	Mathematical CCUS-network models	Yes	Yes	Partial	Limited	Model taxonomy
Present review	Coupled SSM–network planning	Yes	Yes	Yes	Critical evidence appraisal	PRISMA-guided 17-study direct-evidence synthesis + SI extraction matrix

"Partial" indicates that a topic is discussed but is not the organizing focus. The comparison concerns scope and reporting features, not journal quality or citation impact.

source-sink matching and transport network optimization. However, current applications remain largely localized and scenario-specific, and no integrated optimization framework has yet been established to coordinate source-sink matching, transport network design, and operational scheduling throughout the planning and operation process (Gahlot et al., 2025; Li et al., 2025).

Recent reviews provide complementary accounts of SSM progress, CCUS-network modelling, and regional application (Lin et al., 2026; Rahmanta et al., 2026; Table 1). This review addresses a narrower question: how source allocation, shared transport topology and capacity, and operational feasibility are represented jointly in planning models. Its contribution is an explicit evidence framework that distinguishes sequential, jointly optimized, and decomposition-based coordination, and separates demonstrated CCUS results from transferable or prospective AI and digital-twin capabilities. It does not infer a universal method ranking where direct comparative evidence is unavailable. The overall logical structure of this review and the relationships among core concepts are illustrated in Figure 1.

2 Methods

2.1 Review design

This review examines two coupled CCUS infrastructure-planning problems: source–sink matching (SSM) and CO_2 transport-network optimization. It asks how published studies jointly represent source allocation, network topology and capacity, CO_2 flow, and storage availability, and what evidence supports the use of alternative solution methods.

The review defines the scope and evolution of SSM, from point-to-point connections to multi-source, multi-sink, and hub-and-cluster configurations. It then examines transport-network optimization, including transport modes, route and capacity decisions, physical constraints, and governance conditions. The synthesis compares mathematical programming, decomposition, metaheuristics, uncertainty treatment, and emerging data-driven approaches, while distinguishing established applications from prospective methods.

A reproducible database search and screening workflow was used. The Search Methodology subsection reports the databases, search date, search strings, deduplication rule, operational eligibility criteria, screening counts, and evidence-coding protocol (Table 2). The PRISMA 2020 flow diagram is presented in Figure 2; the complete search record, exclusion log, and direct-evidence matrix are provided in the Supplementary Material.

This synthesis contributes an evidence-based comparison of integrated CCUS planning approaches. It defines co-optimization through explicit links among source allocation, network design, operational flow, and storage constraints; distinguishes sequential, monolithic, and decomposition-based coordination; and evaluates methods according to the evidence reported for feasibility, solution quality, computational performance, and predictive or monitoring capability. The cross-study comparison is based on a transparent 17-study direct-evidence subset; the remaining references provide contextual, methodological, physical, governance, or prospective evidence.

2.2 Search methodology

A *de novo* search was conducted in Web of Science Core Collection and Scopus on 16 September 2026. The search combined CCUS/CCS terms with source-sink matching or allocation, CO_2 transport-network or hub/cluster terms, and optimization or planning terms. No year, language, document-type, subject-area, or open-access filters were applied. The complete database-specific strings are reported in Supplementary Table S1a. The search identified 518 Web of Science records and 1,051 Scopus records (n = 1,569 before deduplication).

Records were deduplicated using DOI as the primary identifier; where at least one record lacked a DOI, exact normalized title and publication year were used. A total of 427 duplicates were removed, leaving 1,142 unique records for title-and-abstract screening. Direct-evidence eligibility was defined at the study level and applied to each full-text report. A report qualified only when it (i) examined a CCUS/CCS system containing identifiable CO_2 sources and geological storage sinks; (ii) presented a quantitative planning or

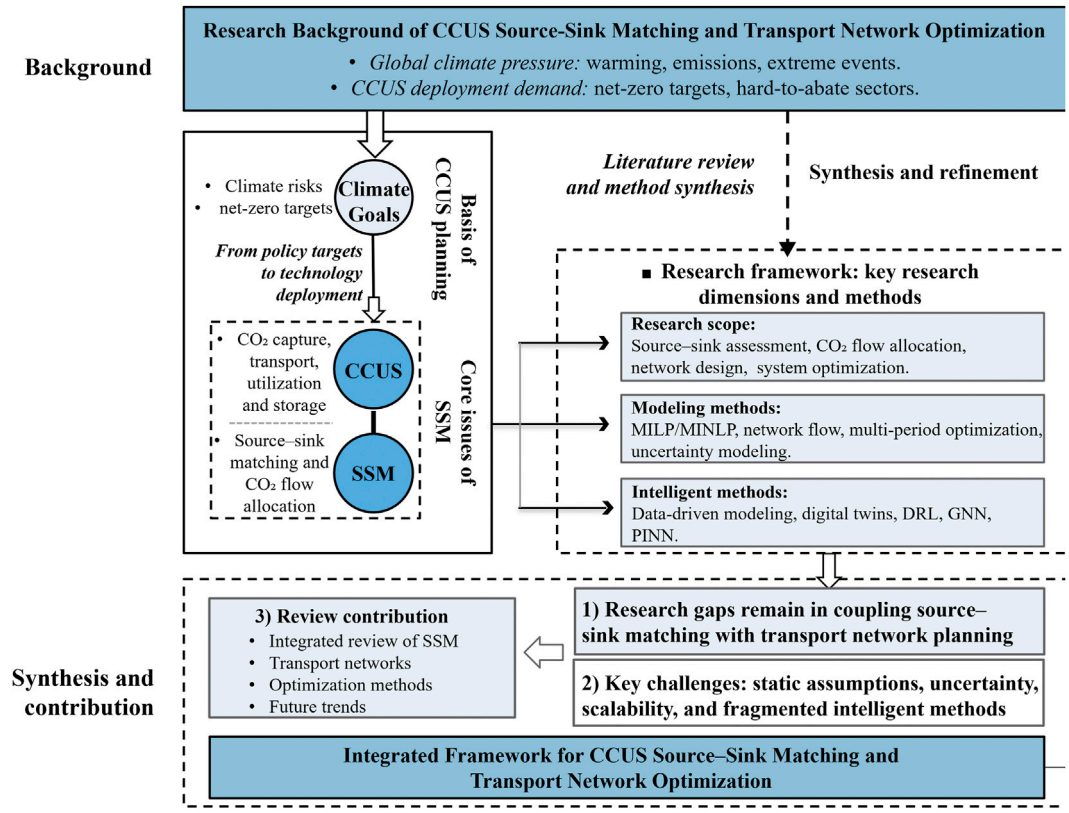


FIGURE 1
Integrated review framework for CCUS source-sink matching and transport network optimization.

TABLE 2 Auditable literature-review protocol.

Protocol element	Operational definition used in this review
Working corpus	1,142 unique database records after removal of 427 duplicates; 17 studies retained for direct evidence synthesis
Targeted update	Web of Science Core Collection and Scopus search on 16 September 2026; 1,569 records before deduplication
Core query	CCUS/CCS AND source-sink matching/allocation or transport-network/hub/cluster AND optimization/planning; database-specific strings in Supplementary Table S1a
Eligibility	Direct CCUS infrastructure-planning relevance; no database search filters; contextual citations distinguished from direct evidence
Extraction	System boundary, planning decisions, formulation, uncertainty treatment, scale, computational reporting, reported outcome, and limitations
Synthesis rule	No effect-size pooling; do not infer missing values; compare methods only on transparent feasibility, objective, runtime, gap, robustness, and training/inference evidence

The protocol documents the revised database-search, screening, and evidence-coding workflow. It does not imply that every cited contextual reference was treated as direct comparative evidence.

optimization model rather than a purely descriptive, screening, or component-performance study; (iii) explicitly linked source–sink allocation to at least one transport-infrastructure decision—such as network topology, route, hub or terminal location, transport mode, capacity, flow, investment timing, or expansion—or represented these decisions jointly within a coupled CCUS supply chain; and (iv) reported extractable results for system configuration, cost, capacity, allocation, flow, deployment, or performance. A study did not need

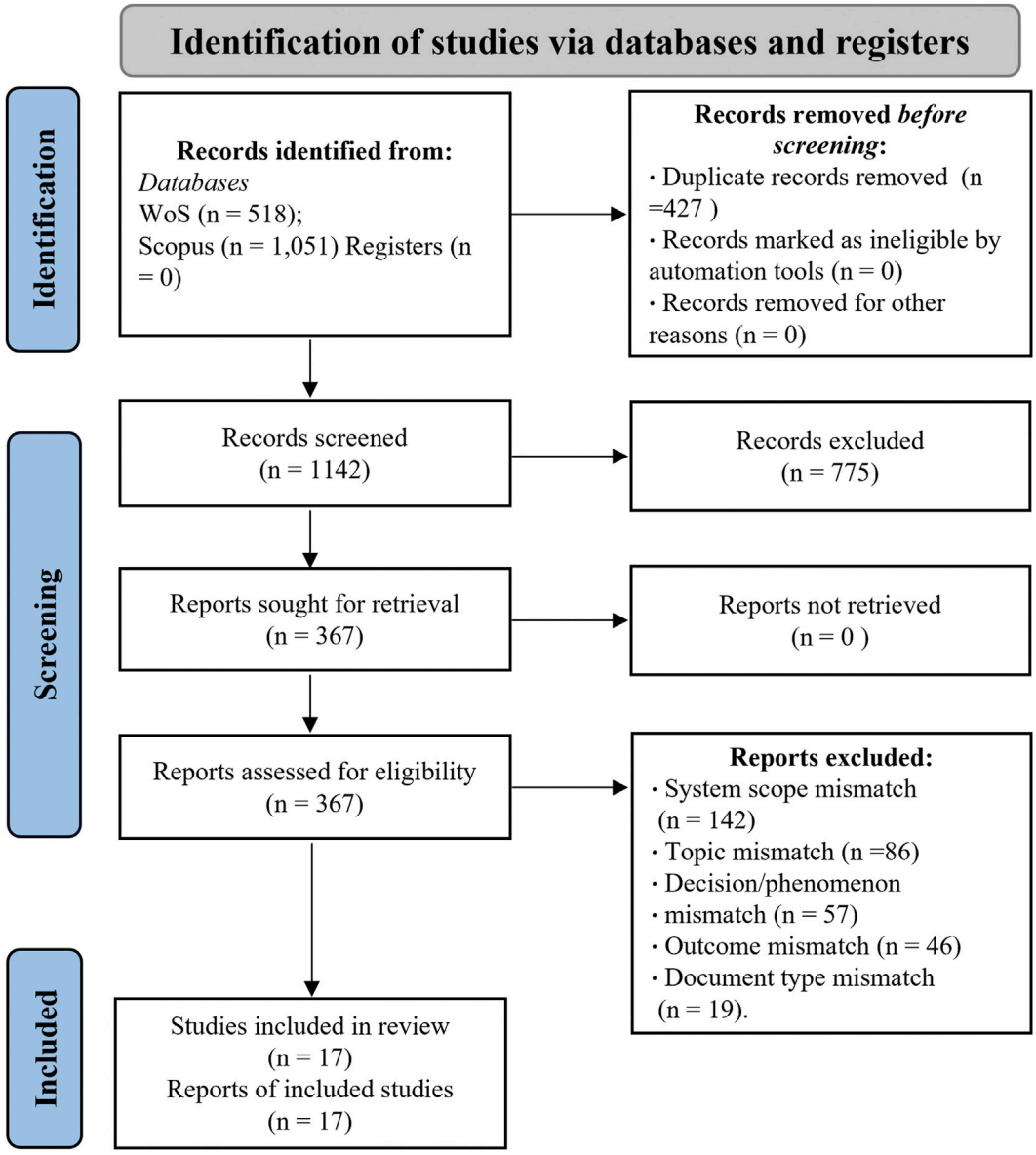


FIGURE 2
PRISMA 2020 flow diagram for identification, screening, eligibility assessment, and inclusion of studies in the direct-evidence synthesis.

to optimize every decision simultaneously; instead, the precise set of endogenous decisions and imposed constraints was coded so that partial and full co-optimization were not treated as equivalent.

The 17 studies were not selected as illustrative or representative cases; they comprise all reports that met the operational eligibility criteria and therefore form the review's direct-evidence subset. Full-text reports were excluded when the CCUS/geological-storage system boundary was absent or only a single component was examined (system-scope mismatch); the paper did not address SSM, transport-network planning, hub/cluster deployment, or a closely coupled supply-chain problem (topic mismatch); no linked allocation, design, capacity, routing, or planning decision was modeled (decision/phenomenon mismatch); the study did not report outcomes that could be extracted for the review questions (outcome mismatch); or the record was an ineligible publication type, such as a review, editorial, protocol, or abstract without a full study (document-type mismatch). Each excluded full-text report was assigned one primary reason to prevent double counting. Papers that informed physical constraints, governance, component technologies, AI/PINN methods, or digital-twin concepts but did not satisfy the direct-evidence definition were retained only as contextual, methodological, or prospective references.

2.3 Study selection and evidence coding

The screening workflow is summarized in Figure 2. Of 1,142 unique records screened, 775 were excluded on title and abstract, and 367 reports underwent full-text assessment. Seventeen studies met all eligibility criteria and were included in the direct-evidence synthesis; 350 full-text reports were excluded for

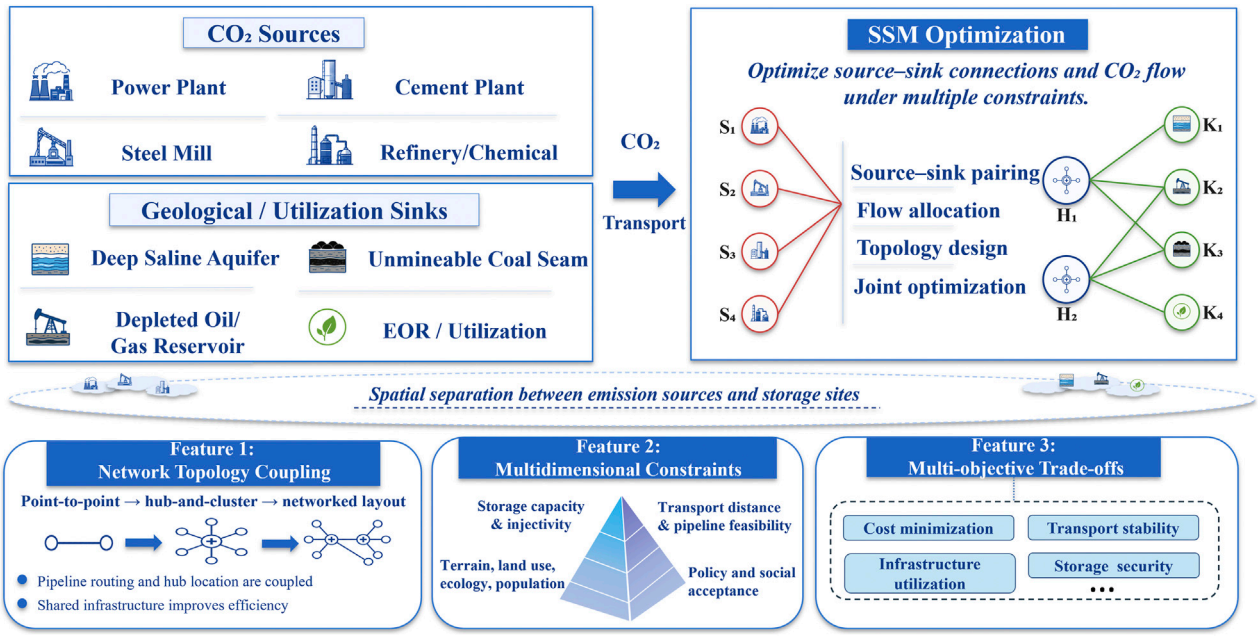


FIGURE 3
Basic concept and key features of CCUS source-sink matching (SSM).

system-scope mismatch (n = 142), topic mismatch (n = 86), decision/phenomenon mismatch (n = 57), outcome mismatch (n = 46), or document-type mismatch (n = 19). Detailed exclusion definitions and the report-level exclusion log are provided in Supplementary Tables S1–S3.

3 Optimization of source-sink matching

Source-sink matching (SSM) is a central decision-making component of the CCUS value chain. Well-designed source-sink configurations are essential for reducing system costs, optimizing storage resource utilization, and enhancing emissions-reduction performance. The complexity of SSM arises from three interrelated challenges. First, the spatial separation of emission sources and storage sites creates strong coupling among network topology, source-sink connectivity, and CO_2 flow allocation. In systems with multiple emission sources and geological storage sites, these interactions give rise to hierarchical network design problems. Second, SSM must simultaneously accommodate geological, engineering, ecological, and policy constraints, including storage capacity, injectivity, transport conditions, and regulatory requirements, which are often strongly interdependent. Third, alternative configurations involve inherent trade-offs among cost, safety, and resilience, potentially generating structural conflicts during decision-making. Consequently, SSM requires the joint optimization of network topology, discrete decisions, nonlinear constraints, and multiple objectives. These combinatorial characteristics render SSM optimization an NP-hard problem that generally cannot be solved in polynomial time. This section reviews the field from three perspectives: fundamental concepts, paradigm evolution, and key technological advances.

3.1 Fundamental concepts of source-sink matching

Within the CCUS value chain, a source is an emission facility with technically feasible CO_2 capture, primarily coal-fired power plants and high-emitting facilities in the iron and steel, cement, chemical, and refining sectors. A sink is a site capable of receiving CO_2 for storage or utilization, including deep saline aquifers, depleted oil and gas reservoirs, unmineable coal seams, and EOR projects (Intergovernmental Panel on Climate Change, 2005; Snæbjörnsdóttir et al., 2020). In multi-source, multi-sink systems, SSM entails the joint optimization of source–sink connectivity, CO_2 flow allocation, and storage-site allocation (Fan et al., 2018; Wang et al., 2026). Specifically, it determines capture volumes, network topology, inter-node flows, and storage volumes within a feasible region defined by geological safety, engineering feasibility, infrastructure availability, policy constraints, and social acceptance. These decisions seek to balance storage security, transport reliability, and full-chain system costs (Becattini et al., 2022; Ji et al., 2026). As the principal interface between capture and storage, SSM determines the spatial organization of CO_2 flows and fundamentally influences both the architecture of the subsequent transport network and the economic performance of the overall system (Fang et al., 2025).

In practice, emission sources and storage sites are rarely co-located because their locations are governed by different structural factors. CO_2-emitting facilities are located primarily according to energy supply, raw-material availability, transport accessibility, and industrial agglomeration. Consequently, they tend to be concentrated in coal-rich regions, heavy industrial bases, port areas, and energy-intensive industrial clusters (Gunawan et al., 2024). Suitable geological formations, by contrast, are determined

by subsurface geological conditions. Deep saline aquifers, depleted oil and gas reservoirs, and unmineable coal seams are therefore concentrated in specific sedimentary basins and oil- and gas-bearing regions (Vangkilde-Pedersen et al., 2009).

The spatial mismatch between emission sources and storage sites is a fundamental source of complexity in SSM. The large number and geographical dispersion of source and sink nodes generate numerous potential connections, creating a high-dimensional combinatorial decision space. Alternative configurations involve different transport distances, infrastructure investments, and storage-utilization efficiencies, while remaining interdependent due to competition for shared resources and constraints on storage capacity (Chen and Hosseini, 2026). SSM must therefore optimize CO_2 flows across multiple sources and sinks at the system level, subject to storage capacity, injectivity, transport distance, pipeline constructability, and system-cost constraints, rather than relying on localized proximity-based matching (Tan et al., 2013; Hasan et al., 2014).

This challenge is further compounded by the high uncertainty and investment irreversibility of CCUS projects. Once a source-sink configuration is selected, it creates substantial lock-in effects by determining the topology of the transport network and the scale of infrastructure investment (Li et al., 2024). Suboptimal matching may lead to system-wide infrastructure inefficiencies or the prolonged underutilization of storage resources. Therefore, long-term expansion requirements should be incorporated at the planning stage to minimize the risk of costly reconstruction or capacity expansion after infrastructure commissioning (Li et al., 2024).

As shown in Figure 3, the spatial mismatch between sources and sinks, together with the need for multi-stakeholder coordination, gives rise to substantial structural complexity in SSM. Rather than being a one-dimensional pairing or route-planning problem, SSM is a combinatorial decision problem in which network topology design, multidimensional constraints, and multi-objective optimization are closely intertwined. This complexity is reflected in three forms of coupling.

The first feature is network-topology coupling. In systems with multiple emission sources and storage sites, these elements, along with hub nodes and transport links, form a complex spatial network. Source-sink connections determine not only CO_2 flow directions but also hub locations, network structures, and transport modes. Although conventional point-to-point connections are structurally simple, they fail to fully exploit infrastructure sharing and economies of scale in large-scale CCUS deployment (Li et al., 2024). In contrast, the hierarchical hub-and-cluster model connects dispersed emission sources to shared infrastructure, thereby increasing pipeline utilization and reducing duplicative construction costs (Sun et al., 2021). For instance, in integrated cement-methanol production systems, CO_2 capture, green hydrogen supply, and methanol synthesis are mutually coupled. Consequently, CO_2 flows form tree-like or networked configurations with multiple nodes and destinations, rather than following a single linear chain (He et al., 2026). Thus, SSM is not merely a pairing problem but a combinatorial optimization challenge involving network topology generation, route selection, and flow allocation (Becattini et al., 2022; Wang et al., 2025).

The second feature is multidimensional constraint coupling. Storage capacity and injectivity constitute fundamental SSM constraints, as they determine both the total CO_2 volume a site can accommodate and its maximum annual injection rate. Early models incorporated source operating lifetimes, storage availability windows, injection rates, and capacities into continuous-time optimization frameworks (Tan et al., 2013). However, practical SSM applications must accommodate broader constraints, including transport distance, compression energy, pipeline capacity, terrain, land use, ecological conservation redlines, population distribution, and regulatory requirements (Wang et al., 2020; Wang et al., 2026). These constraints are strongly interdependent. A geographically proximate storage site may be unsuitable due to limited injectivity or ecological restrictions, whereas a geologically favorable site may be distant, entailing higher transport and infrastructure costs. Similarly, while cross-regional CO_2 transport improves resource utilization and infrastructure sharing, it increases the complexity of network coordination, risk management, and regulatory oversight (Bogs et al., 2026). Therefore, SSM decisions require coordinated consideration of energy-system characteristics, geological conditions, engineering infrastructure, ecological constraints, and the policy environment, rather than relying solely on geological suitability or spatial proximity.

The third feature involves trade-offs among multiple objectives. Early studies primarily sought to minimize transport distance, capital investment, or full-chain system costs to identify the most economically efficient configuration under predefined conditions (Middleton and Bielicki, 2009). However, as CCUS progresses from demonstration projects to large-scale deployment, SSM can no longer be treated solely as a spatial connection problem governed by economic efficiency. It must simultaneously consider storage safety, engineering feasibility, environmental performance, infrastructure utilization, and long-term operational stability (Giakoumi et al., 2026). These objectives often conflict. For example, minimizing transport costs may reduce network redundancy and weaken system resilience, whereas enhancing storage safety may require longer transport distances and greater infrastructure investment. Although increased infrastructure sharing improves resource-use efficiency, it may also complicate regional coordination and operational scheduling (Giakoumi et al., 2026).

These trade-offs become more pronounced in long-term planning due to uncertainties in emitter participation, CO_2 transport demand, policy constraints, carbon prices, and the timing of phased infrastructure development. Excessive dependence on a single storage site or transport corridor may expose the system to stranded assets, transport bottlenecks, and inadequate emissions-reduction capacity if geological conditions change, network nodes fail, or demand shifts (Bogs et al., 2026). Consequently, SSM research increasingly seeks to balance cost reduction with adaptability, network resilience, and long-term robustness across scenarios, thereby supporting CCUS deployment throughout the infrastructure life cycle (Olson and Yaw, 2025).

Accordingly, SSM should be treated as a dynamic system-level optimization problem that jointly represents network topology, interdependent constraints, competing objectives, and long-term uncertainty (Budiarto et al., 2025; Olson and Yaw, 2025).

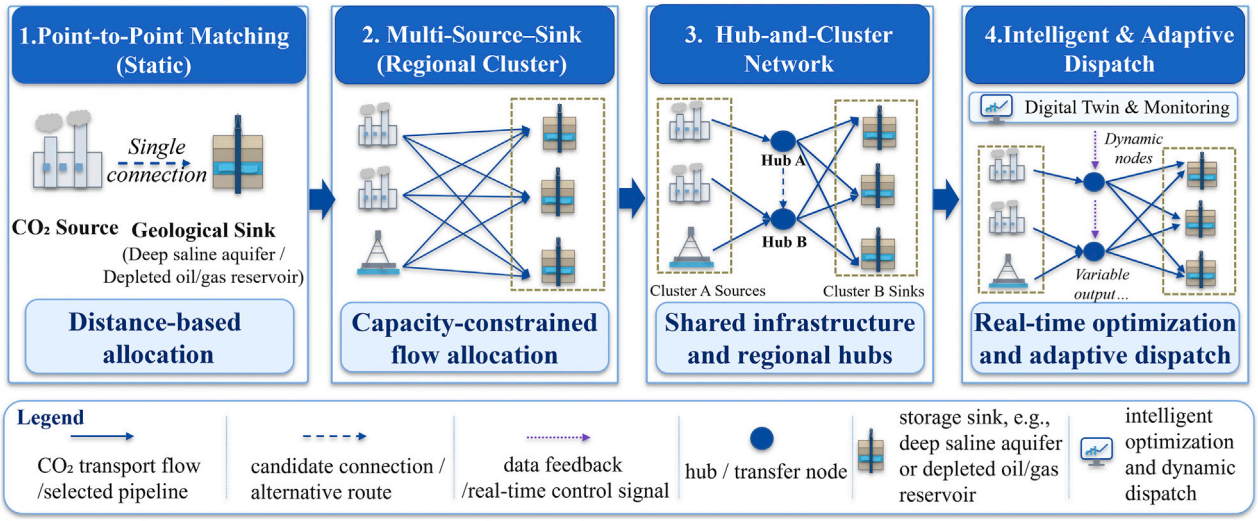


FIGURE 4
Evolution of source-sink matching toward large-scale CCUS deployment.

3.2 Evolution of source-sink matching paradigms

SSM research is increasingly moving beyond the optimization of local pilot projects toward cluster-based deployment across regions, national borders, and ultimately global systems. As illustrated in Figure 4, the dominant paradigm is evolving from simple source-sink connections toward networked, clustered, and dynamic configurations (He et al., 2026; Zhang et al., 2025). Early studies predominantly employed point-to-point matching, pairing individual emission sources with storage sites primarily based on geographical proximity and focusing on the minimization of transport distance or system cost (Boavida et al., 2013).

Hub-and-cluster planning shifts the decision problem from individual source–sink links to shared-capacity design. Hubs aggregate dispersed flows to justify common gathering systems and trunk pipelines; their value therefore depends on phased entry, capacity utilization, third-party access, and cost allocation. Shipping can connect dispersed or early-stage sources to storage or pipeline terminals before dedicated pipelines become economic. Optimization models should consequently determine hub location, trunk and terminal capacity, modal interfaces, connection timing, and flow allocation jointly, rather than treating shared infrastructure as exogenous (Sun et al., 2021; Kattel et al., 2026).

Four capabilities should be distinguished. Dynamic optimization updates allocation, routing, capacity-use, or operating decisions as model states change over time, for example, with changing source availability, storage injectivity, or transport demand (Budiarto et al., 2025). Scenario-based planning evaluates a finite set of plausible future conditions before or during strategic investment decisions; it supports decisions under uncertainty but does not itself require real-time state updates (Bogs et al., 2026; Olson and Yaw, 2025). Monitoring refers to the measurement or estimation of the physical state of capture, transport, and storage assets, including pipeline flow, pressure, and storage behavior (Gahlot et al., 2025; Sleiti et al., 2022). Intelligent decision support uses optimization, simulation, or learning models to interpret available information and recommend actions, but it does not necessarily imply autonomous control. Digital twins may link monitoring and state estimation with decision support. Their use for integrated, closed-loop CCUS network operation remains an emerging research direction rather than an established capability.

The expansion of large-scale SSM has been supported by the development of increasingly comprehensive databases of CO_2 emission sources and storage sites across multiple countries and regions. In Europe, the EU GeoCapacity project systematically evaluated saline-aquifer storage potential in major sedimentary basins, including those in the North Sea region, and established a CO_2 storage database for SSM applications (Vangkilde-Pedersen et al., 2009). In the United States, the Carbon Storage Atlas, developed by the U.S. Department of Energy (DOE) and NETL, integrates data on major stationary CO_2 emission sources and geological storage resources. These data have supported national-scale simulations of CCUS deployment and source-sink matching (National Energy Technology Laboratory, 2015). Similarly, China has developed emission-source databases covering major sectors, including coal-fired power generation, iron and steel, cement, and chemicals, while conducting county-scale surveys and assessments of storage potential in saline aquifers and depleted oil and gas reservoirs (Xiang et al., 2026). In parallel, the Global CCS Institute and other organizations continue to update global CCUS project databases, providing an important data foundation for international and cross-regional SSM research (Minervini et al., 2024).

The evolution of SSM modeling has closely followed the broadening of optimization objectives and decision scales. Early models were typically formulated as MILP problems with the primary objective of minimizing total system costs. Given predefined emission sources, storage sites, and candidate transport routes, these models jointly determined CO_2 capture and storage volumes, source-sink connectivity, pipeline investments, and flow allocation (Middleton and Bielicki, 2009). Their principal advantage lies in the ability to integrate discrete

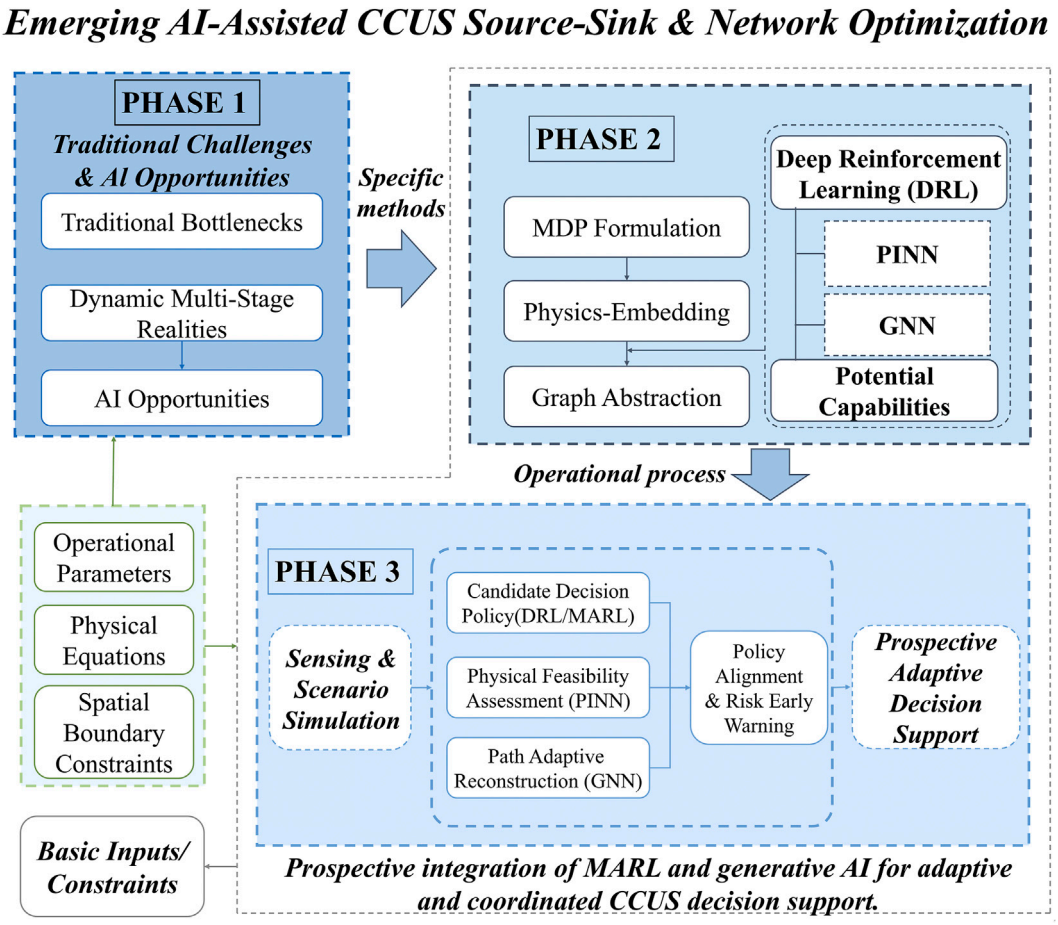


FIGURE 5
AI-enabled methodologies for CCUS source-sink matching and transport network optimization.

decisions, such as facility siting and pipeline construction, with continuous decisions, such as CO_2 flow allocation, within a unified optimization framework. Accordingly, these models have been widely used for the preliminary planning and cost assessment of regional CCUS infrastructure. However, their formulations have generally prioritized economic performance and have represented storage safety, adaptive capacity expansion, intertemporal investment, and uncertainty only to a limited extent (Becattini et al., 2022; Fraga et al., 2021).

Models such as SimCCS integrated geospatial data, candidate-network generation, and mathematical optimization for systems with multiple sources and storage sites (Middleton et al., 2012). Multi-period extensions divide capture, transport, and storage investments among successive stages, representing changes in the source portfolio, phased capacity, and progressive use of storage resources (Budiarto et al., 2025). Integrated supply-chain models broaden the boundary to include capture, compression, transport, utilization, and storage, while scenario, stochastic, and robust formulations seek configurations that remain feasible and economically acceptable under uncertainty in carbon prices, storage conditions, and participation (Becattini et al., 2022; Bogs et al., 2026).

SSM optimization is consequently moving beyond static matching for individual projects toward dynamic planning for large-scale deployment. Research is no longer concerned solely with identifying economically efficient source-sink connections but increasingly addresses the organization and routing of CO_2 flows, infrastructure sharing, cross-regional coordination, and robust long-term operation within integrated networks connecting multiple emission sources and storage sites.

3.3 Key technological advances and emerging methods

3.3.1 Large-scale network modeling and decomposition-based optimization

As CCUS deployment expands, the increasing number of source and sink nodes and the growing complexity of network structures present major challenges for SSM. Large sets of emission sources, storage sites, candidate pipeline corridors, and intermediate nodes create a high-dimensional decision space in which source-sink connectivity, CO_2 flow allocation, pipeline network design, and capacity sizing must be optimized jointly (Sun et al., 2021). Most existing studies address this complexity through a sequential framework comprising candidate network generation, network flow optimization, and capacity configuration. Geospatial data are

first used to identify feasible transport corridors and construct candidate pipeline networks. Mixed-integer programming or network flow models are then applied to determine source-sink connections, CO_2 flows, pipeline investments, and storage allocation (Becattini et al., 2022; Middleton et al., 2012). At national or cross-regional scales, incorporating all emission sources and storage sites into a single integrated model can result in prohibitive computational requirements. Therefore, cluster aggregation, hub selection, and hierarchical solution strategies are widely used to reduce the dimensionality of the problem (Sun et al., 2021). One approach groups geographically proximate sources or facilities with similar industrial characteristics into regional emission clusters and subsequently identifies hubs with substantial storage potential or advantageous transport locations. This reduces the number of candidate source-sink combinations while improving the utilization of shared infrastructure (Sun et al., 2021). An alternative approach employs multi-stage or decomposition frameworks. Interregional CO_2 flows and hub locations are first determined at the strategic level, after which pipeline routes, diameters, and flow allocations are refined at the local engineering level. Such frameworks help reconcile system-wide coordination with project-level implementability (Becattini et al., 2022).

These methods transform SSM from a simple pairing task into a tractable and scalable network design problem, thereby providing an essential methodological basis for national-scale and cross-regional CCUS infrastructure planning.

3.3.2 AI-driven intelligent decision-making methods

AI-related methods should be distinguished by decision horizon and evidence maturity (Table 3). Mathematical programming remains the principal approach for strategic infrastructure planning because it explicitly represents investment, capacity, flow, and feasibility constraints (Kattel et al., 2026). Existing AI/ML reviews focus mainly on materials discovery, process prediction, plume forecasting, and broader system-integration possibilities rather than validated source–sink-network control (Mazhar et al., 2026). Learning-based methods could prospectively support repeated operational decisions when a validated simulation or monitoring environment is available, but the reviewed evidence remains component-level or scenario-specific rather than end-to-end large-scale network deployment.

As illustrated in Figure 5, deep reinforcement learning (DRL) can formulate repeated allocation or scheduling decisions as a Markov decision process and learn policies through interaction with a specified simulation environment (Sutton and Barto, 1998). In this review, this role is presented as a prospective methodological application rather than as direct evidence of operational CCUS network deployment. Its potential role is therefore operational adaptation, rather than strategic network design. Direct demonstrations for large-scale source–sink–network systems and controlled comparisons with exact optimization remain limited. Claims of DRL advantage should report training cost, inference time, feasibility and constraint-violation rates, robustness to distribution shift, and performance against a strong optimization baseline.

Physics-informed neural networks (PINNs) embed governing equations, such as flow, mass conservation, pressure propagation, or storage behaviour, in surrogate or state-estimation models (Raissi et al., 2019; Karniadakis et al., 2021). In CCUS, they may support component simulation, state estimation, or constraint evaluation. However, no common benchmark currently shows that PINNs outperform established optimization models for integrated large-scale source–sink and transport-network design.

Overall, mathematical programming provides the most established basis for auditable strategic planning, whereas metaheuristics remain case-specific search tools. DRL, PINNs, graph learning, generative methods, and digital-twin-enabled decision support are emerging approaches whose end-to-end value for CCUS network planning has not yet been demonstrated on comparable instances. Numerical rankings between AI and conventional optimization are therefore unwarranted unless studies report comparable evidence on feasibility, solution quality or optimality gap, computational budget, robustness, and operational validation.

3.3.3 Digital twin-enabled system resilience optimization

Digital twins and digital shadows can link monitoring data with physics-based models to support state estimation, anomaly detection, and scenario evaluation. In the available literature, CCUS-relevant evidence is limited to component, pipeline, and storage-monitoring applications (Gahlot et al., 2025; Sleiti et al., 2022), whereas broader infrastructure studies provide only transferable architectural context (Bi et al., 2024; Rozhok et al., 2021). These references should not be interpreted as evidence that closed-loop, multi-operator CCUS network optimization has already been operationally validated.

Figure 6 presents the prospective digital-twin-enabled closed-loop architecture considered in this review.

A future digital-twin architecture could combine pipeline and reservoir states, estimated source availability, and a constrained optimization layer to recommend operational recourse within pre-approved strategic capacity bounds. Such a system would require validated sensing, uncertainty calibration, cybersecurity and governance arrangements, human override, and explicit feasibility safeguards; these requirements remain research and deployment priorities rather than demonstrated general capabilities.

Shared hubs and transport infrastructure can create economies of scale, but their benefits depend on utilization, storage availability, contractual access, and phased investment. Digital twins may eventually help monitor these conditions, whereas evidence for real-time network reconfiguration remains prospective.

3.3.4 Limitations and evidence gaps in intelligent methods

Four limitations recur across DRL, PINNs, graph learning, and digital-twin studies. First, representative data on failures, transients,

TABLE 3 Evidence maturity and reporting requirements for conventional and intelligent approaches.

Approach	Typical decision horizon	Guarantee/ certificate	Data or training burden	Comparable CCUS evidence	Current maturity
MILP/MINLP	Strategic planning; some operations	MILP gap and feasibility certificate; nonconvex MINLP depends on solver	Structured parameters; no training	Runtime/gap reported in a minority of studies	Established planning method
Stochastic/robust optimization	Strategic and staged planning	Scenario/robust guarantee under stated uncertainty set	Scenario generation and probability/uncertainty-set design	One robust study reports seconds to 72 h and ≤2% accepted gaps (Bogs et al., 2026)	Research-to-planning
Metaheuristics	Topology and multi-objective search	No global-optimality certificate	Many objective evaluations and tuning runs	Case-specific cost or topology results; runtime/gap often NR	Case-specific
PINN/learned surrogate	Simulation acceleration and state estimation	No discrete-network optimality guarantee	Training collocation/data; architecture and domain calibration	No common full-network benchmark found	Proof-of-concept
DRL/MARL	Dynamic scheduling and control	No general global-optimality guarantee	Large simulated interaction budget; reward/constraint engineering	No controlled CCUS network comparison found	Conceptual/ early
Digital twin/shadow	Monitoring to online support	Depends on embedded estimator, physics and optimizer	Sensors, integration, calibration, governance and cybersecurity	Monitoring demonstrations; closed-loop network evidence scarce	Component-level pilots

The table compares what each method can certify and what the reviewed literature reports. It does not rank methods on incomparable instances. MARL, multi-agent reinforcement learning.

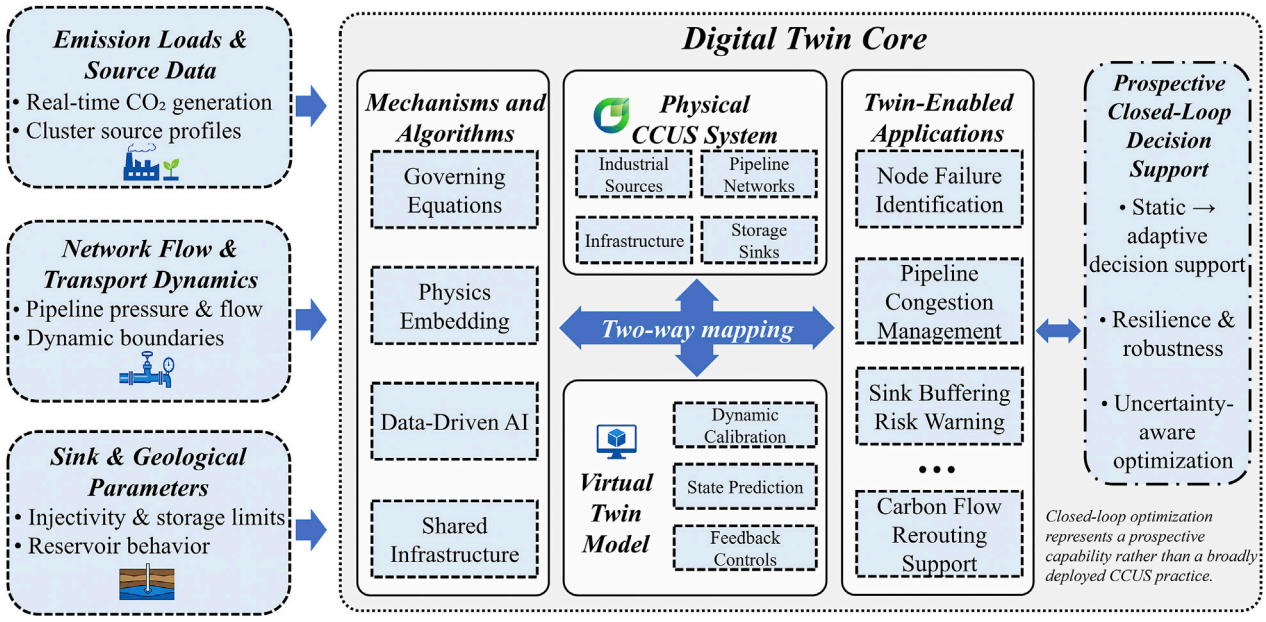


FIGURE 6
Digital twin-enabled closed-loop framework for resilient CCUS networks.

and multi-operator scenarios are scarce; thus, simulation-generated training data inherit model bias. Second, training and calibration may shift computational burdens offline rather than eliminating them, necessitating the reporting of wall-clock inference times alongside training resources. Third, black-box policies and surrogates are difficult to audit in safety-critical permitting processes, particularly when topology or operating conditions diverge from training data. Fourth, differentiable physical penalties do not guarantee satisfaction of mixed-integer, legal, or emergency constraints. Hybrid designs should preserve an

optimization or safety-filter layer to certify feasibility and expose uncertainty rather than returning a single unqualified prediction.

3.4 Integrated source–sink matching and pipeline network Co-optimization

SSM determines source-level capture and allocation decisions, while network design determines whether those allocations can be delivered through feasible corridors and capacities. The two are coupled because a source-sink assignment creates flow on selected arcs, arc construction limits feasible flow, and storage capacity and injectivity limit the destination flow that can be assigned. Treating the stages independently can select apparently short or inexpensive matches that become infeasible or costly after corridor, compression, capacity, or permitting constraints are represented.

3.4.1 Formal definition and coordination architectures

Let q_{ijt} denote CO_2 flow from source i to storage site j in period t, y_{ed} the binary construction decision for link e with capacity option d, and f_{et} the network flow. The model jointly minimizes capture, transport, storage, and infrastructure costs subject to:

$$\sum_j q_{ijt} \le E_{it}, \sum_i q_{ijt} \le \tilde{I}_{jt}, 0 \le f_{et} \le \sum_d \tilde{F}_{ed} y_{ed}$$

Terminal balances link allocation to network flows:

$$a_{it} = \sum_j q_{ijt}, b_{jt} = \sum_i q_{ijt}$$

A model is co-optimized only when allocation, infrastructure, and operational decisions are solved jointly, or coordinated while preserving system-wide feasibility and a common objective (Middleton and Bielicki, 2009; Hasan et al., 2014).

Sequential feedback iterates between SSM and network design but may remain sensitive to initialization and does not generally guarantee a system-wide optimum. Monolithic MILP/MINLP formulations jointly optimize assignment, topology, diameter, flow, and timing, although binary decisions and nonlinear hydraulics can limit tractability. Decomposition provides an intermediate architecture: a master problem can select sources, sinks, corridors, and capacities, while operational or scenario subproblems test flow, injection, hydraulic, and uncertainty feasibility and return cuts or coordination signals.

Decomposition methods partition strategic integer decisions from operational, hydraulic and scenario-dependent subproblems. In Benders decomposition, the master problem determines sources, sinks, corridors and capacities, while subproblems validate operational feasibility and parameter uncertainty, returning cutting planes to update master solutions. ADMM coordinates distributed regional subproblems through boundary flows and capacity pricing, suiting decentralized data environments. However, standard ADMM convergence guarantees fail under nonconvex hydraulic constraints and discrete decisions, necessitating rigorous reporting of primal feasibility, residuals and centralized benchmark gaps. All coordination architectures should be compared on identical test cases using total cost, constraint violations, runtime, optimality gaps and partitioning sensitivity as key metrics.

3.4.2 International deployment evidence

Operating and planned hubs reveal constraints that abstract network models often omit. Porthos combines a shared onshore gathering system with an offshore pipeline and depleted-gas-field storage, illustrating the value of open-access trunk capacity and staged emitter connection (Porthos, 2026). Northern Lights uses ships to aggregate geographically dispersed emitters before transferring them through a subsea pipeline to offshore storage; its phased expansion makes multimodal scheduling and capacity reservation central decisions (Equinor, 2026). In North America, the Summit Carbon Solutions proposal demonstrates that a technically connected network can still be reshaped by permitting, landowner, and routing constraints; the Iowa filing covered approximately 688 miles of 6–24 inch pipeline (Iowa Utilities Commission, 2026). These cases motivate the endogenous treatment of phasing, access rules, multimodal interfaces, and route approval (Table 4).

4 CO_2 transport network optimization

CO_2 transport and network design are critical to the economic viability and emissions-reduction efficiency of CCUS systems. While SSM determines source-sink connectivity and CO_2 flow allocation, transport network optimization addresses transport modes, network topology, pipeline routing, pipe diameters, pressure ratings, and capacity allocation. SSM provides the volume and spatial distribution of CO_2 flows for network design, whereas transport costs and engineering feasibility inform source-sink decisions. As CCUS transitions from demonstration projects to large-scale deployment, transport systems are evolving from isolated pipeline configurations toward multimodal coordination, shared networks, and dynamic optimization. This section reviews transport modes, key factors influencing route planning, optimization models and solution methods, and major future challenges in CO_2 transport network optimization.

4.1 Comparison of CO_2 transport modes

Transport-mode selection is fundamental to optimizing CO_2 transport systems, with economic performance and applicability largely dependent on transport volume, distance, and spatial distribution (Gunawan et al., 2024). Current CCUS transport systems are dominated by pipelines, supplemented by shipping, rail, and road transport, thereby forming a multimodal system suited to diverse application scenarios (Uddin et al., 2025), as illustrated in Figure 7.

Pipelines dominate CO_2 transport by supporting large-scale, continuous, and stable flows. They are particularly suitable for long-term connections among major emission sources, regional CO_2 hubs, and geological storage sites. As throughput and utilization increase, pipelines reduce unit transport costs and achieve substantial economies of scale (Becattini et al., 2022). However,

TABLE 4 Representative deployment and model scales relevant to integrated planning.

Case/study	Region and status (audit date)	Sources/sinks	Reported transport scale	Planning implication
Porthos (2026)	Rotterdam–North Sea; construction in progress	Multiple port emitters/ depleted gas fields	>30 km onshore + >20 km offshore; avg. 2.5 Mt a^{-1}; 37 Mt storage	Shared trunk capacity and phased access
Northern Lights (Equinor, 2026)	Norway/ cross-border; Phase 1 completed	Multiple ship-delivered emitters/offshore reservoir	1.5 Mt a^{-1} Phase 1; ~100 km subsea; expansion to ≥5 Mt a^{-1}	Ship–hub–pipeline coordination and reservation
Summit Carbon Solutions (Iowa Utilities Commission, 2026)	US Midwest; proposed/ permitting	Dispersed biogenic sources/ geological storage	Iowa filing: ~688 miles, 6–24 inch	Route approval and stakeholder constraints
Li et al. (2024)	China; planning model	1,186 sources/329 sinks	~80,700 km candidate/shared layout	National-scale geospatial and combinatorial burden

Project status was checked on 4 August 2026. Project figures describe different system boundaries and are not cost-performance benchmarks. DN, nominal diameter.

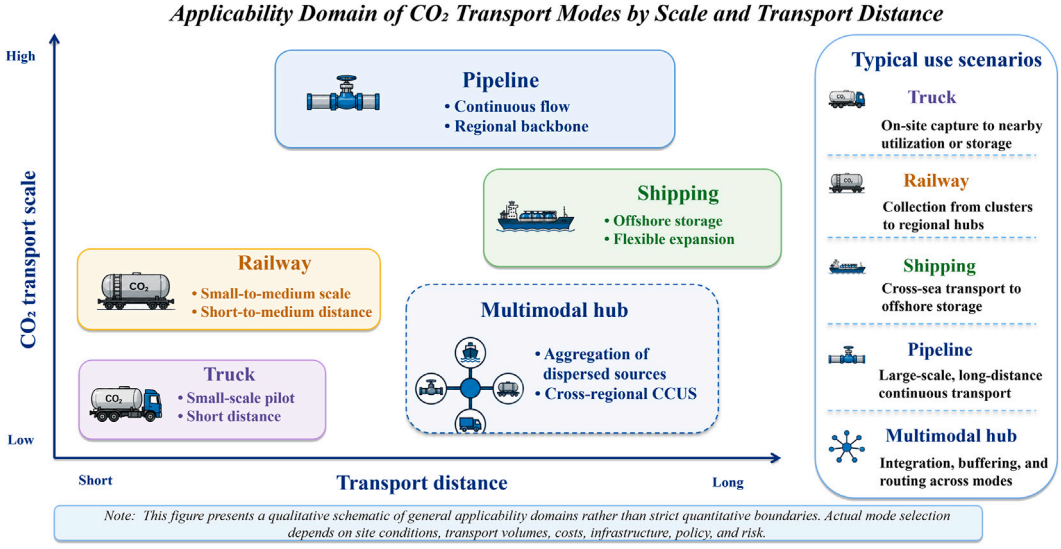


FIGURE 7
Applicability map of CO_2 transport modes.

their high upfront capital requirements make economic performance highly dependent on transport volume and network utilization. Consequently, pipelines are best suited to regions with concentrated emission sources, stable demand, and clearly identified storage sites (Bogs et al., 2026).

Shipping, rail, and road transport are more appropriate for low-volume flows, dispersed sources, and early-stage demonstration projects (Al Baroudi et al., 2021). Shipping offers flexible routing and relatively low initial investment, making it suitable for offshore and long-distance transport or for regions lacking established pipeline networks (Global CCS Institute, 2025c). Coastal shipping demand has also been studied in a broader transport context (Schrobback et al., 2024). Rail and road transport are better suited to short-distance, small-scale, or temporary CO_2 transfer; however, their limited carrying capacity, service frequency, and higher unit costs restrict their use for continuous large-scale transport (Uddin et al., 2025). Therefore, while pipelines remain preferable in concentrated, high-throughput settings, dispersed or complex cross-regional systems may require multimodal configurations combining pipelines with shipping, rail, or road transport.

CO_2 gathering hubs can integrate dispersed emission sources and connect pipelines with rail or shipping to form multimodal transport networks, thereby improving system flexibility while reducing overall transport costs (Hyypiä et al., 2026). In hard-to-abate sectors and cross-regional deployments, multimodal transport and coordinated regional optimization have become important means of enhancing CCUS system resilience (Kattel et al., 2026). Table 5 compares the applicable conditions, cost structures, key technical parameters, and typical applications of different CO_2 transport modes. Pipelines are most suitable for large-scale continuous transport, shipping for cross-regional transport and offshore storage, and rail and road transport for small-scale or

TABLE 5 Comparison of CO_2 transport modes for CCUS deployment.

Mode	Indicative application	Cost structure	Physical/interface requirements	Reported cost evidence	Typical role
Pipeline	Stable, high-throughput corridors; medium/long distance	High up-front CAPEX; strong utilization economies	Dense/supercritical phase; pressure, impurities, fracture and compressor design are project-specific	0.25–55.82 EUR t^{-1} across reported designs at 1,000–25,000 t d^{-1} (Solomon et al., 2024)	Cluster gathering, shared trunks, storage connection
Shipping	Dispersed sources, offshore or cross-border routes, staged volumes	Vessels, terminals and liquefaction; often OPEX-sensitive	Liquid CO_2 specification; buffers, berthing and ship–pipeline scheduling	USD 9–50 t^{-1} across selected 1–5 Mt a^{-1}, 600–1,200 km project studies (Global CCS Institute, 2025c)	Flexible aggregation and offshore access
Rail	Small/medium inland flows or transition stage	Tank cars, loading terminals, dispatch and handling	Batch logistics and terminal storage; specification is chain-specific	NR on a common CCUS network basis	Isolated emitters and temporary links
Truck	Small, short-distance, pilot or contingency flows	Vehicle/driver/fuel dominated; repeated loading	Road access, pressure/liquid tanks and local safety controls	NR on a common CCUS network basis	Pilots, sampling and emergency logistics

CAPEX, capital expenditure; OPEX, operating expenditure; vol% = volume percent; ppm = parts per million.

demonstration projects. Thus, the four modes are strongly complementary (Uddin et al., 2025; Gunawan et al., 2024).

4.2 Factors affecting pipeline network design and route planning

Pipeline network optimization necessitates the joint design of network structure and operating strategies. Existing studies primarily address four dimensions: spatial and economic constraints, network topology, physical and safety conditions, and system coordination with social constraints.

1. Spatial and economic constraints

Pipeline routing depends not only on source-sink distance but also on terrain, land use, existing transport and energy corridors, construction difficulty, land acquisition, and operating costs (Gunawan et al., 2024). The shortest route does not necessarily minimize total system costs. Crossing densely populated areas, ecologically sensitive zones, or complex terrain can substantially increase construction, maintenance, and risk-management costs (Gunawan et al., 2024). Consequently, pipeline planning must balance transport distance, capital investment, construction accessibility, and operating efficiency (Bogs et al., 2026).

2. Network topology

Routing and network-design methods differ in their representation and evidential strength. MILP and network-flow models jointly select corridors, diameters, and flows under discrete investment and linearized capacity constraints, and can provide optimality bounds for a stated formulation. MINLP models add pressure, hydraulic, or phase constraints, but their scalability and solution certification are more limited. Metaheuristic and bio-inspired methods can explore nonconvex topology and routing alternatives, but require repeated-run statistics, feasibility checks, and optimization benchmarks (Kattel et al., 2026; Wei et al., 2026). Point-to-point networks suit early projects, whereas trunk-and-branch and hub-and-spoke structures support shared capacity at larger scale (Sun et al., 2021; Kryeziu et al., 2025).

3. Physical and safety conditions

CO_2 pipeline transport typically occurs under dense-phase or supercritical conditions, requiring careful control of pressure, temperature, pressure drop, and phase stability (Osiadacz et al., 2026). Impurities such as water, H_2S, and SO_x can cause corrosion, hydrate formation, and flow instability, thereby affecting material selection, compressor-station placement, and operating parameters (Sun et al., 2022). Leakage dispersion, CO_2 accumulation in low-lying areas, and emergency-response capacity also influence pipeline routing and risk-control measures (Al Baroudi et al., 2021). Therefore, pipeline planning must evaluate flow stability, material integrity, and leakage risk alongside economic performance.

4. System integration and social constraints

Governance affects network design when it defines modelable conditions, including permitted corridors, access rules, tariffs, cost-sharing, capacity reservation, liability costs, and participation thresholds. These factors can enter optimization models as constraints, costs, or scenarios. By contrast, public acceptance (Witte, 2021), regulatory timing, and inter-operator negotiation are usually better represented as exogenous scenarios or eligibility conditions unless empirical parameters are available. Existing studies seldom endogenize these institutional variables, although they can determine whether technically feasible shared infrastructure is investable and implementable (Bogs et al., 2026; Global CCS Institute, 2025a; Schomakers et al., 2025).

4.2.1 Mathematical modeling framework

The essence of CCUS pipeline network optimization is the joint optimization of network structure and operating strategies under multiple constraints. Figure 8 illustrates a typical graph-theoretic representation of a CCUS pipeline network. Let V denote the set of

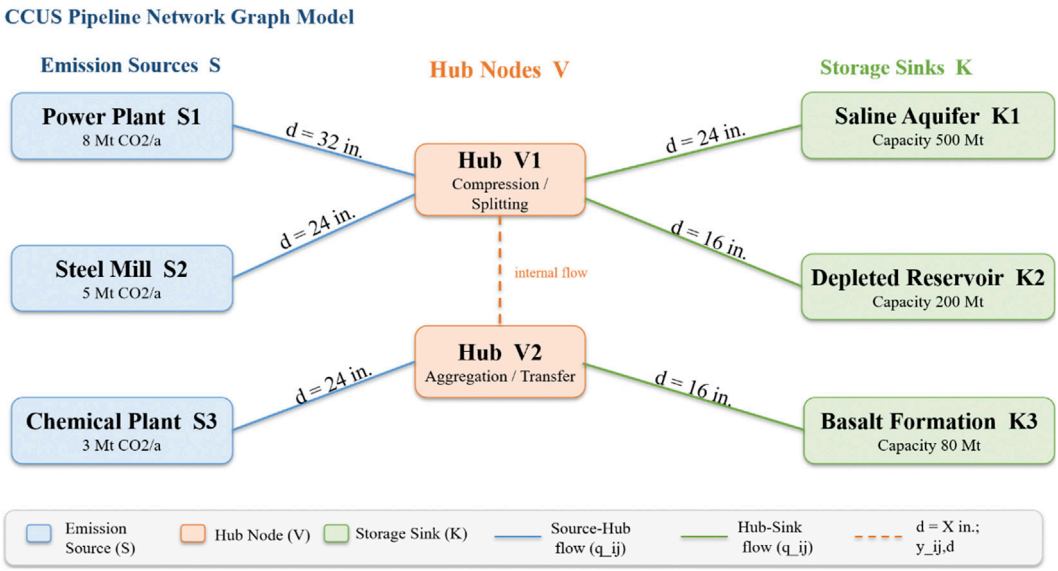


FIGURE 8
CCUS pipeline network graph model.

nodes, including emission sources S and storage sites K; A denote the set of candidate pipeline arcs; and D denote the set of available pipe diameters. In the network, source nodes, such as power plants and industrial facilities, are connected to intermediate hubs or storage sites through directed arcs. Each arc represents a potential pipeline with a set of candidate diameter options, while CO_2 flows between nodes satisfy flow-conservation constraints. This graph-theoretic model captures both the discrete selection of network topology and the continuous allocation of CO_2 flows, providing a unified framework for subsequent mathematical optimization.

CCUS pipeline network models typically integrate integer decisions, such as pipeline construction and diameter selection, with continuous decisions, such as flow allocation. When the objective function and constraints are linearizable, the problem is formulated as a mixed-integer linear programming (MILP) model, with some studies seeking Pareto-optimal solutions for multiple objectives (Budiarto et al., 2025). Conversely, the inclusion of nonlinear pressure, hydraulic, or phase-behavior constraints results in a mixed-integer nonlinear programming (MINLP) formulation.

Existing modeling frameworks differ primarily in their optimization objectives. Early studies focused on minimizing full-chain system costs, encompassing capital and operating expenses for capture, transport, and storage (Hasan et al., 2014). Later models incorporated the maximization of CO_2 emissions reductions to align with policy-driven decarbonization goals (Middleton et al., 2012). Multi-objective models further balance system cost, CO_2 handling capacity, and infrastructure utilization, whereas uncertainty-oriented models minimize regret or worst-case costs (Bogs et al., 2026; Olson and Yaw, 2025). Additionally, some integrated supply-chain models include life-cycle carbon emissions or the levelized cost of abatement (LCOA) to evaluate system performance under varying policy scenarios (Becattini et al., 2022; Budiarto et al., 2025).

Among these formulations, minimizing the discounted full-chain system cost remains the most prevalent objective in MILP models. Consequently, this study adopts it as the representative objective function (Hasan et al., 2014).

1. Basic formulation.

$$\min Z = \sum_{(i,j)\in A} \sum_{d\in D} c_{ij,d}^{\mathrm{cap}} y_{ij,d} + \sum_{(i,j)\in A} c_{ij}^{\mathrm{op}} q_{ij} + \sum_{i\in S} c_i^{\mathrm{capt}} x_i + \sum_{k\in K} c_k^{\mathrm{stor}} q_k$$

here, $y_{ij,d} \in \{0,1\}$ is a binary decision variable indicating whether diameter d is selected for pipeline segment (i, j); $q_{ij} \geq 0$ denotes the corresponding CO_2 flow rate (t/a); $x_i \in \{0,1\}$ indicates whether a capture facility is installed at emission source i; $q_k \geq 0$ denotes the annual injection volume at storage site k (t/a); and c^{cap}, c^{op}, and c^{stor} are the coefficients for capital, operating, and storage costs, respectively. The objective function incorporates the main costs of CO_2 capture, transport, and storage. Its basic structure originates from the SimCCS model (Middleton and Bielicki, 2009), which was later extended by Hasan et al. (2014) to include investment in capture facilities (Hasan et al., 2014). It has since been widely applied to CCUS source-sink matching, transport network design, and storage system planning (Kattel et al., 2026).

The constraints mainly include the following categories:

1. Flow-conservation: For each intermediate hub node $v \in \mathcal{V}/(\mathcal{S} \cup \mathcal{K})$, the total CO_2 inflow equals the total outflow (D'Amore et al., 2018).

$$\sum_{i:(i,v)\in A} q_{iv} = \sum_{j:(v,j)\in A} q_{vj}, \quad \forall v \in$$

2. Capture capacity: The CO_2 output from emission source i cannot exceed its maximum capture capacity Q_i^{max} (Hasan et al., 2014).

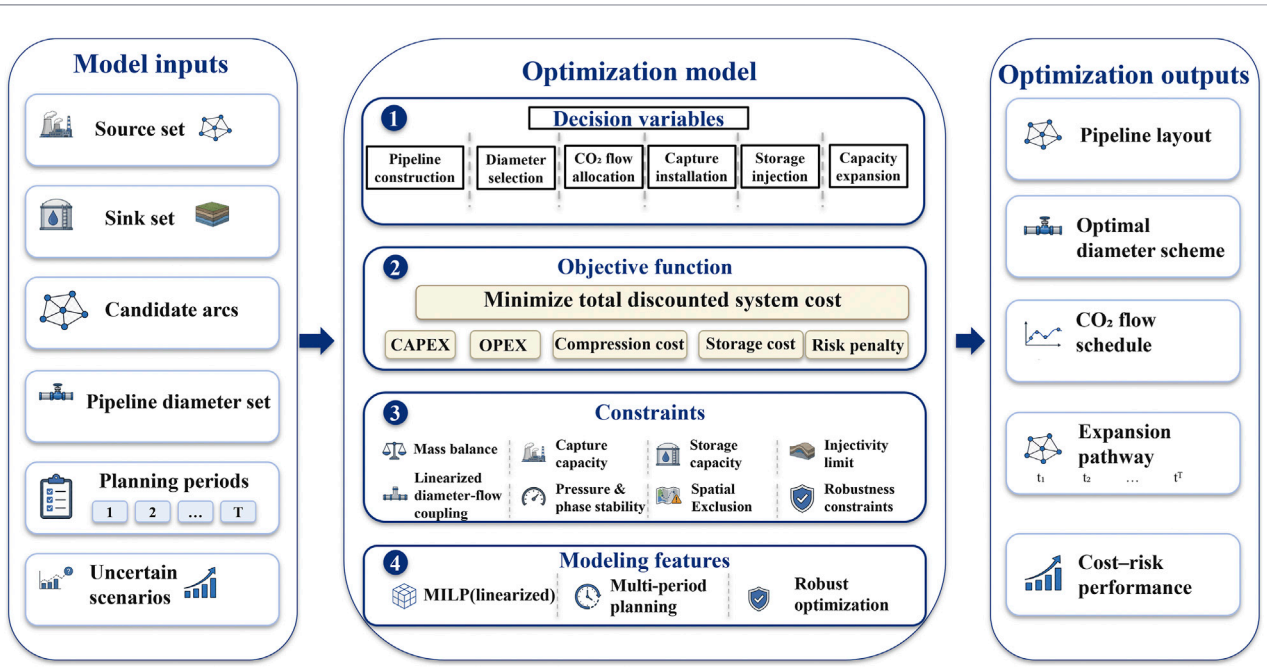


FIGURE 9
Mathematical formulation of the MILP-based CO_2 pipeline network optimization model.

$$\sum_{j:(i,j)\in A} q_{ij} \le Q_i^{\max} x_i, \quad \forall i \in S$$

3. Storage capacity: For each storage site $k \in \mathcal{K}$, cumulative CO_2 injection cannot exceed the available storage capacity C_k (Sun et al., 2021).

$$q_k \le C_k, \quad \forall k \in K$$

4. Flow-diameter coupling: The flow through each pipeline segment must remain within the capacity of the selected diameter and satisfy the operating pressure range for supercritical CO_2 transport, $[P^{\min}, P^{\max}]$ (Solomon et al., 2024).

$$\sum_{d\in D} Q_{ij,d}^{\min} y_{ij,d} \le q_{ij} \le \sum_{d\in D} Q_{ij,d}^{\max} y_{ij,d}, \quad \forall (i,j) \in A$$

5. Diameter selection: Each candidate pipeline segment can be assigned at most one diameter (D'Amore et al., 2018).

$$\sum_{d\in D} y_{ij,d} \le 1, \quad \forall (i,j) \in A$$

In engineering applications, the model must also account for operating pressure, pressure drop, phase stability, and impurity effects. To retain MILP tractability, these physical constraints are commonly represented through piecewise linearization, predefined capacities for each pipe diameter, or hydraulic approximations (Sleiti et al., 2022; Osiadacz et al., 2026). In multi-period planning, the model is extended by introducing a time index $t \in T$ and a discount factor δ^t to represent investment timing and discounted capital costs. Middleton and Bielicki (2009) first developed this dynamic framework in the multi-period version of SimCCS (Middleton and Bielicki, 2009), and Fraga et al. (2021) later formalized it as a multi-period MILP with a new-construction variable Δy:

$$min\, Z = \sum_{t\in T} \delta^t \left[\sum_{(i,j)\in A} \sum_{d\in D} c_{ij,d}^{\text{cap}} \cdot \Delta y_{ij,d}^t + \sum_{(i,j)\in A} c_{ij}^{\text{op}} \cdot q_{ij}^t + \sum_{k\in K} c_k^{\text{stor}} \cdot q_k^t \right]$$

here, $\Delta y_{ij,d}^t \in \{0,1\}$ indicates whether pipeline segment (i,j) with diameter d is newly constructed or expanded in period t. The variables q_{ij}^t and q_k^t denote the pipeline flow and storage-site injection volume in period t, respectively, while δ^t is the corresponding discount factor. The new-construction variable prevents capital investment from being counted repeatedly in subsequent periods. This dynamic framework has been applied to model the phased deployment of CCUS infrastructure (Wu et al., 2025).

Under robust optimization, uncertain parameters are represented by an uncertainty set, and the model minimizes the worst-case cost over that set (Bogs et al., 2026). Unlike stochastic programming, it does not require probability distributions. Multi-stage formulations further introduce recourse decisions, allowing capacity, routing, or operation to be revised as uncertainty is resolved.

$$\min_{x\in X} \max_{\xi\in\mathcal{U}} C(x,\xi).$$

These models enhance the robustness of pipeline network designs against uncertain emitter participation, storage availability, and long-term demand fluctuations (Bogs et al., 2026; Olson and Yaw, 2025). Overall, MILP and its extensions constitute

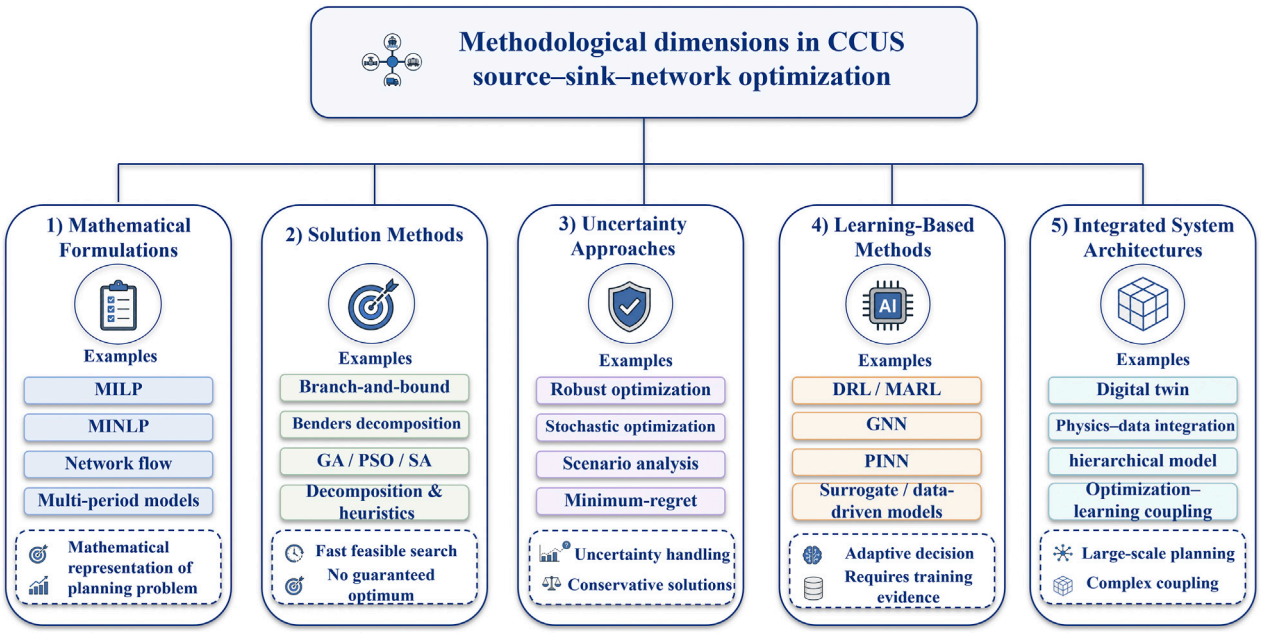


FIGURE 10
Taxonomy of optimization methods for CCUS source-sink-network planning.

the mathematical foundation of CCUS pipeline network optimization. Figure 9 summarizes the model inputs, optimization process, and outputs. Current models are evolving from static deterministic planning toward multi-period dynamic planning, robust optimization, and resilient network design.

4.2.2 Solution methods

Figure 10 presents the classification of optimization approaches used in this review.

Solution approaches include exact optimization, decomposition, metaheuristics, and learning-assisted simulation or control. Table 6 reports only study-specific quantitative results. Runtime, cost, network scale, and optimality gaps are not directly comparable across studies because hardware, solver settings, model scope, physical representation, data, and stopping criteria differ. Accordingly, unavailable metrics are reported as NR, and no method ranking is inferred from these values.

1. Classical mathematical programming methods. Mixed-integer linear programming (MILP) is among the most widely used methods for CCUS pipeline network optimization. Binary variables represent discrete decisions, such as pipeline construction, diameter selection, and facility siting, while continuous variables describe CO_2 flows and operational levels. These models can jointly address network topology, capacity constraints, flow balance, and system costs (Tapia et al., 2018). Mixed-integer nonlinear programming (MINLP) is employed when nonlinear processes, such as pressure loss, fluid flow, and compressor operation, are modeled explicitly, although this substantially increases computational complexity (Bogs et al., 2026).

For large networks, branch-and-bound and decomposition methods, such as Benders decomposition, column generation, and Lagrangian relaxation, can improve solution efficiency by exploiting problem structure (Kryeziu et al., 2025). More recently, mathematical programming has been integrated with robust optimization and stochastic programming to address uncertainty in emitter participation, storage capacity, and economic parameters (Kattel et al., 2026). Related stochastic methods have been reviewed for natural-gas pipelines (Wu et al., 2018).

2. Metaheuristic optimization methods. Large-scale CCUS pipeline network optimization is often nonlinear, high-dimensional, mixed-integer, and multi-objective, which limits the applicability of exact algorithms to large instances. Consequently, metaheuristics such as genetic algorithms (GA), particle swarm optimization (PSO), and simulated annealing (SA) are widely used to obtain high-quality near-optimal solutions at lower computational cost (Rahmanta et al., 2026; Ríos-Mercado and Borraz-Sánchez, 2015). These methods do not require convexity or gradient information and are well suited to complex search spaces, network topology design, multi-objective Pareto optimization, and MINLP problems (Coello Coello, 2006; Deb, 2011). Recent studies have combined metaheuristics with mathematical programming, surrogate models, and machine learning to improve computational efficiency and solution robustness (Rahmanta et al., 2026). Additionally, bio-inspired approaches have simulated tree-like preferential flow in porous media to generate feasible pipeline topologies with lower infrastructure costs (Solomon et al., 2024).
3. AI-enabled optimization and modeling methods. DRL, PINNs, and digital twins have distinct potential roles. DRL

TABLE 6 Reported scale and computational evidence for representative optimization studies.

Study/method	Case scale	Solver/ hardware	Runtime	Gap/accuracy evidence	Interpretation
Middleton and Bielicki (2009), SimCCS MILP	Regional scalable source–sink network	NR	NR	Exact MILP framework; gap NR	Foundational integrated infrastructure model
Hasan et al. (2014), MILP	US nationwide, regional and state cases	NR	NR	50%–80% stationary-source reduction scenarios; gap NR	Broad supply-chain scope; computational metadata incomplete
Li et al. (2024), GIS + optimization	1,186 sources; 329 sinks; ~80,700 km	NR	NR	Network-scale output; gap NR	Large national case, but no transferable runtime boundary
Promsuk et al. (2025), clustering + AHA	Thailand; four candidate sinks	NR	NR	No global-optimality certificate	Case-specific metaheuristic result
Bogs et al. (2026), robust MILP	German instances and regret approximations	Gurobi 12; 36 cores; 96 GB	5 s–258,900 s (72 h cap)	1% target; ≤2% accepted; approximation errors 0.30%–2.90%, one regret case 20.97%	Strongest transparent compute reporting in reviewed set

Values are transcribed at the study level and are not directly comparable. NR, means not reported in the accessible record; AHA, artificial hummingbird algorithm. The Bogs et al. study also reported mean piecewise-linear approximation errors of 10.09%, 1.55%, and 0.52% for one, two, and three segments, respectively, with average runtimes of 33 min, 14 h, and 23 h.

may support repeated operational allocation or scheduling in a validated simulation environment, rather than replace strategic network design (Mazhar et al., 2026). PINNs can support component flow simulation, pressure prediction, or state estimation by embedding physical equations in surrogate models (Karniadakis et al., 2021; Zhai et al., 2023; Ajayi et al., 2026). Digital twins may link monitoring data with simulation and constrained decision support (Sleiti et al., 2022). Evidence for integrated, large-scale CCUS network deployment remains limited; claims should therefore report training effort, inference time, feasibility, and robustness against optimization baselines.

4. Hierarchical decomposition strategies. Hierarchical decomposition is increasingly used for large interregional and global-scale networks. This approach first allocates CO_2 flows among regions at the macro level and then refines local pipeline designs at the micro level. Such a top-down strategy decomposes large optimization problems into tractable subproblems, thereby reducing computational complexity (Zhang et al., 2025).

No single method dominates across strategic design, staged expansion, and real-time operation. Exact formulations remain valuable for auditable feasibility and gap information; decomposition exploits separability; metaheuristics broaden the searchable topology space but typically lack global guarantees; and learned components can reduce repeated simulation costs after training. Hybrid methods should be evaluated using identical data, constraints, and hardware, reporting both offline and online computational effort.

4.3 Challenges in CO_2 transport network optimization

As CCUS transitions from demonstration projects to large-scale deployment, transport-network optimization evolves into a decision-making process that integrates source-sink matching (SSM), corridor and capacity planning, and system operation. The primary challenges include model complexity and physical fidelity, multiple sources of uncertainty, and the coordination of competing objectives alongside heterogeneous data (Figure 11).

1. Trade-off between model complexity and computational efficiency. CO_2 pipeline optimization entails coupled decisions regarding source-sink connections, diameter selection, compressor scheduling, and flow allocation, involving both continuous and integer variables. Incorporating phase stability, pressure drop, and thermodynamic constraints often transforms a mixed-integer linear programming (MILP) model into a nonconvex mixed-integer nonlinear programming (MINLP) problem, substantially increasing computational difficulty. Nonlinear thermodynamic equations and compressor performance curves create highly nonconvex formulations, rendering global optimization of large networks computationally demanding. Consequently, linear approximations, predefined pipeline capacities, and simplified hydraulic constraints are commonly employed to improve tractability (Sleiti et al., 2022; Osiadacz et al., 2026). At the national scale, networks connecting over one thousand emission sources and hundreds of storage sites must also account for geospatial connectivity, sectoral competition, and cost heterogeneity, further intensifying the trade-off between model scale and computational efficiency (Xiang et al., 2026).
2. Robustness under multiple sources of uncertainty. CCUS network planning is exposed to uncertainty in CO_2 supply, storage capacity and injectivity, carbon prices, policy, and disruption. Scenario analysis tests plausible futures but does not optimize recourse. Stochastic programming evaluates expected performance across probabilistic scenarios, whereas robust and minimum-regret models protect against adverse realizations. Multi-period adaptive models further revise investment, capacity, routing, or operation as information arrives. Current

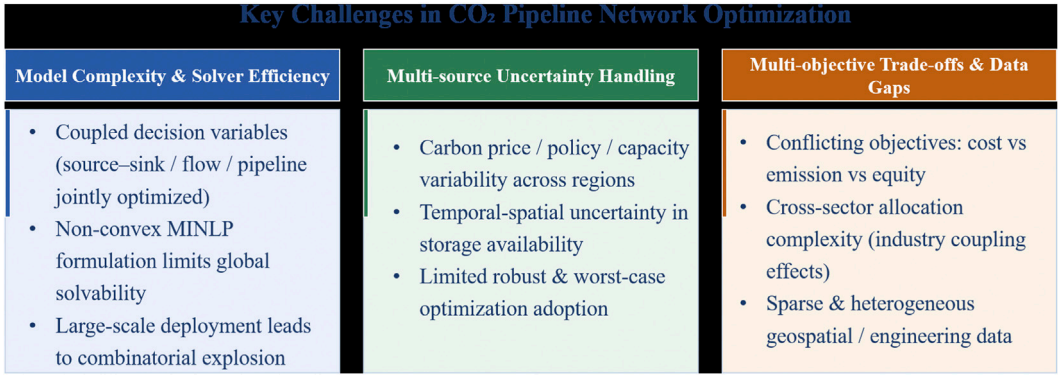


FIGURE 11
Key challenges in CO_2 pipeline network optimization.

studies mainly treat individual uncertainties; coupled uncertainty, disruption recovery, and information-dependent recourse remain limited (Budiarto et al., 2025; Bogs et al., 2026; Olson and Yaw, 2025). Future comparisons should report expected and worst-case cost, regret, stranded capacity, and service shortfall.

3. Multi-objective coordination and heterogeneous data integration. Pipeline optimization must balance system cost, emissions-reduction efficiency, infrastructure utilization, and interregional equity, despite structural conflicts among these objectives. Cost heterogeneity and cross-sectoral externalities further complicate cost and benefit allocation along the value chain. Furthermore, integrating geological, engineering, and policy data is hindered by inconsistent data quality, mismatched spatial and temporal scales, and limited cross-scenario transferability, thereby reducing model fidelity to real-world systems (Hu et al., 2026).

Future CO_2 transport network optimization must therefore address high-dimensional model solutions, decision-making under uncertainty, multi-objective coordination, and heterogeneous data integration. Models should evolve from offline planning toward intelligent, dynamic, and life-cycle decision-making while balancing computational efficiency, solution quality, and physical feasibility.

5 Discussion

This discussion addresses three questions. First, what is established? The evidence supports treating source–sink allocation, transport-network design, and storage use as coupled infrastructure decisions, with mathematical programming providing the clearest feasibility and optimality evidence for a stated formulation. Second, what remains uncertain? No universal method ranking is supported because studies differ in model scope, physical representation, data, and computational reporting. Third, what methodological development is required? Progress depends on common benchmarks, explicit physical-feasibility checks, uncertainty-aware staged planning, and governance-aware coordination. The following sections examine these issues from methodological, demand-side, and supply-side perspectives.

5.1 Exact optimization vs. metaheuristics vs. AI-driven methods

The principal methodological disagreement concerns what constitutes a credible solution for a large, uncertain, and safety-critical CCUS network. The three schools are not mutually exclusive, but they prioritize different forms of evidence: exact optimization emphasizes formal guarantees, metaheuristics emphasize flexible search in difficult design spaces, and AI-driven methods emphasize rapid adaptation. Their reported results are often incomparable because studies use different boundaries, spatial resolutions, hydraulic approximations, hardware, stopping rules, and training budgets. The controversy is therefore as much about evaluation standards as about algorithms.

Mathematical programming provides auditable strategic decisions because variables, constraints, and objective trade-offs are explicit; solver bounds apply, however, only to the formulated model (Middleton et al., 2012; Kattel et al., 2026; Bogs et al., 2026). Linearized hydraulics, predefined corridors, discrete capacity options, and limited scenarios improve tractability but may omit pressure, phase, or transient-flow conditions that affect engineering feasibility. Conversely, detailed thermodynamics, nonlinear hydraulics, multi-stage investment, and extensive uncertainty scenarios can make large instances intractable. Studies should therefore report model fidelity separately from solver optimality, including hydraulic or mass-balance residuals, phase-limit violations, feasibility tolerances, runtime, and optimality gaps.

The metaheuristic school argues that evolutionary and bio-inspired search can accommodate nonlinear objectives, multiple criteria, and topology choices without requiring convexity or a tractable algebraic form (Deb, 2011; Wei et al., 2026). Such methods are useful for generating diverse designs and screening broad candidate spaces, especially when each design is assessed by simulation. Their weakness is evidential: performance can depend strongly on population size, parameter tuning, random seeds, repair

operators, and stopping rules, while a feasible solution provides no general bound on missed improvement. Comparisons based on a single run or against a weak baseline can therefore exaggerate gains. Repeated runs, fixed evaluation budgets, feasibility rates, and comparisons with exact bounds on smaller instances are essential.

The AI-driven school emphasizes DRL, graph learning, PINNs, surrogates, and digital twins for repeated or online decisions in changing environments (Fang et al., 2025; Gahlot et al., 2025; Mazhar et al., 2026; Sleiti et al., 2022). Once trained, these methods can reduce inference time and assimilate operational data, but training cost is frequently excluded from runtime claims. Black-box policies may violate mixed-integer, legal, or emergency constraints; distribution shift can also invalidate performance when emitters, storage properties, or topology differ from the training set. Physics-informed losses improve consistency but do not by themselves certify feasibility. Claims of superior adaptability should therefore be supported by out-of-distribution tests, uncertainty calibration, constraint-violation reporting, and a transparent safety or optimization layer.

The appropriate method consequently depends on the decision horizon and assurance requirement. Exact or decomposition-based optimization is preferable for infrequent strategic investments when cost recovery, permitting, and capacity commitments require traceable decisions. Metaheuristics are most defensible for exploratory multi-objective design or nonconvex topology search when repeated-run statistics and feasibility repair are reported. Learned policies are better suited to repeated operational recourse after a validated strategic design, particularly when latency matters and state data are available. The most credible trend is hybrid rather than substitutive: metaheuristics or learned surrogates generate candidates, exact models certify or repair them, and digital twins update states. Common benchmark instances should report end-to-end time, hardware, training effort, objective bounds, constraint violations, and robustness under matched scenarios.

5.2 Demand-side challenges

Large-scale CCUS deployment remains constrained by uneven abatement incentives across steel, cement, chemicals, and power generation. In many jurisdictions, carbon prices and market coverage do not offset project capital and operating costs, so deployment still depends on subsidies or demonstration support (IEA, 2025; Jiang et al., 2020; Global CCS Institute, 2024). Network models should therefore represent participation thresholds and policy-dependent demand rather than assume that all technically capturable emissions enter the system. The future source portfolio is also uncertain: retirements, electrification, and industrial restructuring will change the location, volume, and operating lifetime of captured CO_2. Fixed-source SSM can therefore lock infrastructure into declining demand. Multi-period models should endogenize emitter entry and exit, ramp-up, and optional capacity expansion (Budiarto et al., 2025; Olson and Yaw, 2025). Interregional coordination is a further demand-side constraint. Spatial mismatch between industrial sources and onshore or offshore storage creates long-distance and cross-jurisdictional flows, while differences in carbon markets, ownership, and benefit allocation hinder shared infrastructure (Yiallourides et al., 2026). Co-optimization should therefore include access rules, cost-sharing, and regional participation rather than treat coordination as an exogenous policy assumption. CO_2 utilization provides only a limited and product-dependent sink relative to geological storage. Models should avoid assuming stable large-scale utilization demand and instead distinguish market-constrained conversion pathways from durable storage, with scenario-dependent prices, capacities, and emissions accounting (Hepburn et al., 2019).

5.3 Supply-side challenges

On the supply side, future CCUS development faces persistent constraints in technological capability, infrastructure availability, and institutional support.

First, model fidelity competes with computational tractability. National-scale networks combine many nodes, candidate corridors, capacity blocks, stages, and uncertainty scenarios. MILP/MINLP and heuristic approaches can represent subsets of these decisions, but their scalability should be judged using identical instances, explicit feasibility tolerances, wall-clock time, and optimality evidence (Kattel et al., 2026; Middleton et al., 2012). Second, shared infrastructure remains limited, sustaining duplicated assets, low utilization, and high transport costs (Global CCS Institute, 2024; Li et al., 2024). Hub-and-cluster and multimodal systems can improve economies of scale, but open-access capacity, phased connection, and cost recovery must be modeled explicitly rather than assumed (Sun et al., 2021). Third, digital twins, DRL, GNNs, and PINNs remain predominantly at validation or scenario-specific stages. Engineering deployment requires interoperable data, calibrated uncertainty, auditable constraints, and evidence that online decisions remain safe outside the training domain (Gahlot et al., 2025; Mazhar et al., 2026). Fourth, inconsistent rules for pipeline access, storage liability, risk sharing, monitoring, and decommissioning impede cross-regional systems (Yao et al., 2025). Life-cycle standards and credible carbon-pricing, finance, and liability arrangements are therefore model inputs, not merely implementation context, because they determine participation, capacity reservation, and long-term cost recovery. In summary, demand creation and supply capacity must be coordinated. Carbon-market incentives, emitter participation, shared infrastructure, regulation, and digital assurance jointly determine whether a technically optimized network is investable and operable. This coupling motivates a shift from static project selection toward staged regional planning and life-cycle decision support.

5.4 Potential developments and emerging paradigms

Over the next decade, digital-twin-enabled optimization is expected to move from proof-of-concept monitoring toward bounded operational decision support. The transition will require more than faster data feeds: physical models, sensor data, and

optimization states must share traceable interfaces, while uncertainty calibration and a safety filter define which actions may be automated. Early digital-shadow and pipeline-twin studies provide components of this architecture, but not yet full closed-loop assurance (Gahlot et al., 2025; Sleiti et al., 2022).

Multi-agent reinforcement learning is likely to emerge where emitters, transport operators, and storage providers retain separate information and objectives. Agents could coordinate capacity reservation, routing, and injection through shared prices or constraints, extending agent-based multimodal and Markov-game approaches (Uddin et al., 2025; Chen and Hosseini, 2026). Progress will depend on constrained learning, privacy-preserving coordination, stable behavior under entry or failure, and comparison with centralized optimization; unconstrained reward maximization is not adequate for safety-critical operation.

Carbon-market-driven network evolution is expected to become a central planning paradigm as carbon prices, contracts, tax credits, and third-party-access tariffs affect when emitters connect and whether shared capacity is bankable. Future models will likely integrate market clearing, cooperative cost allocation, and infrastructure expansion across scenarios rather than treating carbon price as a single exogenous trajectory. Evaluation should distinguish efficient network design from distributive outcomes by reporting utilization, stranded capacity, cost recovery, and stakeholder-specific gains (Bogs et al., 2026; Wang et al., 2026; Yiallourides et al., 2026).

Integration with hydrogen and CCS hubs may transform corridor planning by coupling CO_2 capture and transport with hydrogen supply, synthetic-fuel production, ports, and shared energy infrastructure. Cement–methanol co-production illustrates potential cross-vector synergies, while hub studies show the importance of aggregation and staged connections (He et al., 2026; Sun et al., 2021). These synergies should not be assumed: joint models must account for incompatible operating profiles, energy penalties, additional dependencies, and alternative uses of scarce low-carbon hydrogen.

Quantum-assisted optimization is the most speculative direction. No study in the audited 86-reference corpus demonstrates a quantum advantage for a realistic CCUS network. A defensible near-term research program would therefore test hybrid quantum–classical routines only on clearly defined subproblems—such as candidate-arc selection or scenario sampling—and compare solution quality, total elapsed time, scaling, and energy use with strong classical solvers. Forecasts should remain conditional until hardware limits, encoding overhead, and reproducibility are addressed.

Overall, the reviewed evidence supports coupled planning of source allocation, transport capacity, and storage use, with mathematical programming providing the most auditable basis for strategic decisions. It does not establish a general advantage for metaheuristics, AI, or digital twins, because comparable evidence on solution quality, computational effort, and operational robustness remains limited. Priority advances include physically valid large-scale benchmarks, adaptive planning under uncertainty, and governance-aware shared-infrastructure models. The five research questions below translate these needs into measurable evaluation priorities.

5.5 Priority research questions and evaluation priorities

RQ1. How can physically valid co-optimization be scaled to national systems? Existing studies screen large source–sink sets or model selected capacity and injectivity risks, but do not jointly demonstrate scale, discrete expansion, and dense-phase hydraulics (Li et al., 2024; Xiang et al., 2026; Kattel et al., 2026). Benchmarks exceeding 1,000 sources and 300 sinks should report feasibility, hydraulic residuals, phase violations, runtime, memory, optimality gaps, and sensitivity to hydraulic approximations.

RQ2. How should adaptive investment respond to uncertainty and disruption? Multi-period, robust, and minimum-regret models capture selected uncertainties but rarely link new information to recourse and multimodal recovery (Budiarto et al., 2025; Bogs et al., 2026; Olson and Yaw, 2025). Matched scenario trees should compare expected and worst-case cost, regret, stranded capacity, service shortfall, recovery time, and value of information against static and perfect-information baselines.

RQ3. Which governance mechanisms can sustain shared, multi-owner CCUS networks? Hub and cross-border studies highlight shared capacity but generally treat tariffs, access, and cost allocation as exogenous (Sun et al., 2021; Uddin et al., 2025; Yiallourides et al., 2026). Candidate mechanisms should be assessed for budget balance, individual rationality, coalition stability, incentive compatibility, utilization, cost recovery, emissions served, and distributional outcomes.

RQ4. How should AI-enabled methods be benchmarked against optimization baselines? DRL, PINNs, graph learning, and surrogates show case-specific promise, but no common CCUS protocol accounts for training, inference, and feasibility repair (Fang et al., 2025; Gahlot et al., 2025; Karniadakis et al., 2021; Mazhar et al., 2026). Studies should report hardware, data, seeds, total computational effort, feasible-solution rates, repair time, objective gaps, calibration, and out-of-distribution performance.

RQ5. What evidence is required before digital twins can support closed-loop decisions? Existing digital shadows and pipeline twins support monitoring and state estimation but do not establish safe multi-operator control (Gahlot et al., 2025; Sleiti et al., 2022). Validation should cover sensing and latency, estimation error, uncertainty calibration, fault detection, system uptime and interoperability, human override, auditability, and degradation under missing or corrupted data.

Data availability statement

Publicly available datasets were analyzed in this study. This data can be found here: All data were extracted from previously published peer-reviewed journal articles and conference proceedings cited in the reference list. No new data were generated by the authors, and no centralized data repository or accession numbers are associated with this study. The original

publications can be accessed through Web of Science, Scopus, Engineering Village, and Google Scholar.

Author contributions

DH: Writing – original draft, Writing – review and editing. XW: Writing – review and editing, Writing – original draft. LY: Writing – review and editing. TZ: Writing – review and editing, Writing – original draft.

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The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. During the preparation of this manuscript, the authors used ChatGPT (OpenAI; GPT-5 model; web version accessed 12 August 2026; https://chatgpt.com) for language editing and manuscript-format conversion. The authors reviewed and edited the output and take full responsibility for the content of this publication.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fenvs.2026.1966624/full#supplementary-material

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